

Category 1: The Core Basics (I/O, Variables, Math Operators)

1. Simple Unit Converter

- **Description:** A CLI tool that converts a value from one unit to another (e.g., kilometers to miles, Celsius to Fahrenheit). It asks the user for the value and the conversion type.
- **Core Concepts:**
 - **Input/Output:** Getting the value and conversion choice from the user.
 - **Variables:** Storing the value, choice, and result.
 - **String Formatting:** Printing a clear, formatted result like `"10 km is equal to 6.21 miles."`
 - **Operators:** Using multiplication, division, addition, and subtraction for the formulas.
 - **If/Elif/Else:** To select the correct conversion formula based on the user's choice.

2. Bill Splitter & Tipper

- **Description:** Asks for the total bill amount, the number of people, and the tip percentage. It then calculates the tip amount, the total amount, and how much each person should pay.
- **Core Concepts:**
 - **Input/Output:** Gathering bill, people, and tip percentage.
 - **Variables:** Storing all the numerical values.
 - **Operators:** Division (`/`), multiplication (`*`), addition (`+`).
 - **String Formatting:** Clearly presenting the final breakdown, rounding to two decimal places (`.2f`).
 - **User-Defined Functions:** Create a function `calculate_split(bill, people, tip_percent)` that returns the amount per person.

3. Simple Interest Calculator

- **Description:** Calculates simple interest based on the principal amount, rate of interest, and time period.
- **Core Concepts:**
 - **Input/Output:** Getting principal, rate, and time.
 - **Variables:** Storing the inputs.
 - **Operators:** Multiplication (*), division (/).
 - **String Formatting:** Displaying the calculated interest and the new total amount.
 - **User-Defined Functions:** A function `simple_interest(p, r, t)` to perform the calculation.

4. Time Duration Calculator

- **Description:** Asks the user for two time durations in hours, minutes, and seconds. It then calculates and displays the total sum of the two durations.
- **Core Concepts:**
 - **Input/Output:** Taking 6 separate integer inputs.
 - **Variables:** Storing hours, minutes, and seconds for both durations.
 - **Operators:** Modulo (%) and integer division (//) are key here to handle the carry-over of seconds to minutes and minutes to hours.
 - **Conditional Statements:** To check if seconds are ≥ 60 or minutes are ≥ 60 and adjust the values accordingly.

5. Area & Perimeter Calculator

- **Description:** A menu-driven app that asks the user which shape they want to calculate for (e.g., 1 for Circle, 2 for Rectangle, 3 for Square). It then asks for the required dimensions and outputs the area and perimeter.
- **Core Concepts:**
 - **If/Elif/Else:** To handle the user's menu choice.
 - **Nested If:** Not needed here, but a good place to start thinking about it.

- **User-Defined Functions:** Create separate functions like `circle_area(radius)`, `rectangle_perimeter(length, width)`, etc.
 - **String Formatting:** To display the final results clearly.
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Category 2: Mastering String Formatting & User Functions

6. Mad Libs Story Generator

- **Description:** The program prompts the user for a list of words (e.g., a noun, a verb, an adjective, a celebrity's name) and then inserts them into a pre-written story template.
- **Core Concepts:**
 - **Input/Output:** Gathering multiple string inputs.
 - **Variables:** Storing each word.
 - **String Formatting:** Using f-strings or the `.format()` method to inject the words into the story.
 - **User-Defined Function:** A function that takes all the words as arguments and returns the final story string.

7. Username & Password Generator

- **Description:** Asks the user for their first name, last name, and birth year. It then generates a unique username (e.g., `j.doe1990`) and a simple but strong password suggestion.
- **Core Concepts:**
 - **String Formatting:** Combining strings (concatenation), slicing strings (e.g., `first_name[0]`).
 - **Operators:** String repetition (`*`) could be used for password padding.

- **User-Defined Functions:** A function to generate the username and another for the password.

8. Text Analyzer

- **Description:** Takes a sentence or a short paragraph as input and provides simple statistics, like the number of characters, number of words, and the average word length.
- **Core Concepts:**
 - **Input/Output:** Getting a multi-word string.
 - **String Methods:** Using `.split()` to get words, `.replace()` to clean up.
 - **Operators:** Division (`/`) to calculate the average.
 - **User-Defined Function:** `analyze_text(text)` that returns a dictionary or a formatted string with the results.

9. Simple Encryption/Decryption Tool

- **Description:** A basic "Caesar Cipher" tool. It asks the user for a message and a shift key (a number). It can either encrypt or decrypt the message by shifting each letter.
- **Core Concepts:**
 - **If/Elif/Else:** To choose between 'encrypt' and 'decrypt'.
 - **Nested If:** Inside the loop (if you use one), to check if a character is a letter and if it's uppercase or lowercase.
 - **Operators:** Addition and modulo (`%`) for the shifting logic.
 - **User-Defined Functions:** `encrypt(message, key)` and `decrypt(message, key)`.

10. Initials Extractor

- **Description:** Asks the user for their full name and extracts the initials. For example, "John Fitzgerald Kennedy" becomes "J.F.K."
- **Core Concepts:**

- **Input/Output:** Getting a full name.
 - **String Formatting & Methods:** Using `.split()`, `.upper()`, and `.join()`.
 - **User-Defined Function:** `get_initials(full_name)`.
 - **Conditional Statements:** To handle cases where the user might not enter a middle name.
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Category 3: Diving into Conditional Logic (if/elif/else, Logical Operators)

11. Movie Ticket Pricer

- **Description:** Calculates the price of a movie ticket based on the user's age and the time of the day. (e.g., Children under 12 get a discount, seniors over 65 get a discount, and matinee shows are cheaper).
- **Core Concepts:**
 - **If/Elif/Else:** To handle the different pricing tiers.
 - **Logical Operators (and, or):** To check for conditions like `age < 12 or age > 65`.
 - **User-Defined Function:** `get_ticket_price(age, show_time)`.

12. Grade Calculator

- **Description:** Asks the user for a numerical score (0-100) and converts it to a letter grade (A, B, C, D, F) based on a standard grading scale.
- **Core Concepts:**
 - **If/Elif/Else:** The perfect use case for this. `if score >= 90: grade = 'A', etc.`
 - **Logical Statements:** `if score >= 80 and score < 90:.`
 - **Conditional Statements:** This is the core of the app.

13. BMI Calculator with Health Advice

- **Description:** Calculates the user's Body Mass Index (BMI) from their weight and height. It then provides a health category (Underweight, Normal, Overweight, Obese).
- **Core Concepts:**
 - **Operators:** Division and multiplication for the BMI formula.
 - **If/Elif/Else:** To determine the category based on BMI ranges.
 - **String Formatting:** To display the BMI and the corresponding advice.

14. Leap Year Checker

- **Description:** Asks the user for a year and tells them if it's a leap year or not.
- **Core Concepts:**
 - **Logical Operators:** The logic is a classic: `(year % 4 == 0 and year % 100 != 0) or (year % 400 == 0)`.
 - **Modulo Operator (%):** Essential for this task.
 - **If/Else:** To give the final "Yes" or "No" answer.
 - **User-Defined Function:** `is_leap_year(year)` that returns `True` or `False`.

15. Simple Login System

- **Description:** A program with a hardcoded username and password. It prompts the user to enter their credentials and grants or denies access.
 - **Core Concepts:**
 - **If/Else:** To check if the entered credentials match.
 - **Logical Operators (and):** `if entered_user == "admin" and entered_pass == "1234":`.
 - **Nested If:** A good opportunity to use it if you want to give specific error messages (e.g., "Username not found" vs. "Incorrect password").
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Category 4: Advanced Combinations (Nested If, Functions, All Concepts)

16. Rock, Paper, Scissors Game

- **Description:** The user plays a single round of Rock, Paper, Scissors against the computer.
- **Core Concepts:**
 - **If/Elif/Else:** To determine the winner based on the rules.
 - **Nested If:** You can structure it as `if user_choice == 'rock': if computer_choice == 'scissors': ...`
 - **User-Defined Functions:** A function to get the computer's choice and another to determine the winner.
 - **Random Module:** (A slight extension, but highly recommended) `import random` to make the computer's choice.

17. Advanced Calculator

- **Description:** A CLI calculator that can perform addition, subtraction, multiplication, and division. It should handle invalid input (like dividing by zero) gracefully.
- **Core Concepts:**
 - **If/Elif/Else:** To select the operation.
 - **Nested If:** To check for division by zero inside the 'division' block.
 - **User-Defined Functions:** Separate functions for each operation (`add(a, b)`, `divide(a, b)`).
 - **Conditional Statements:** To check the user's operator input.

18. Guessing Game (Hot/Cold)

- **Description:** The program "thinks" of a number between 1 and 100. The user tries to guess it. After each guess, the program says "Too high," "Too low," or

"Correct!" It also gives "hot/cold" hints based on how close the guess is to the actual number.

- **Core Concepts:**

- **Loops:** (A necessary extension for this to work, but simple `while` loops are easy to learn here).
- **If/Elif/Else:** To compare the guess to the secret number.
- **Nested If:** To provide "hot/cold" hints (e.g., `if guess > secret: if (guess - secret) < 10: print("Hot!")`).
- **Logical Operators:** To check for winning conditions.

19. Restaurant Ordering System

- **Description:** A simple menu is displayed. The user can order items by entering a number. The program calculates the total bill, applies taxes, and can add a tip.
- **Core Concepts:**
 - **If/Elif/Else:** For the menu selection.
 - **User-Defined Functions:** `display_menu()`, `get_item_price(item_number)`.
 - **Operators:** For calculating totals and taxes.
 - **String Formatting:** To display a final receipt.

20. Zodiac Sign Finder

- **Description:** Asks the user for their birth month and day. It then determines their zodiac sign.
- **Core Concepts:**
 - **Nested If:** This is a perfect use case. You can structure it as: `if month == 1: if day < 20: sign = 'Capricorn' else: sign = 'Aquarius'.`
 - **Logical Operators:** For cases that span two months: `if (month == 3 and day >= 21) or (month == 4 and day <= 19):`.
 - **User-Defined Function:** `get_zodiac(month, day)`.

21. Basic Adventure Game (Text-Based)

- **Description:** A short, choice-based adventure. "You are in a dark room. Do you (1) go left or (2) go right?" The user's choices lead to different outcomes (win, lose, or another choice).
- **Core Concepts:**
 - **Nested If:** The core of the game's branching logic. The user's first choice leads to another `if/elif/else` block.
 - **String Formatting:** To describe the scenes and outcomes.
 - **User-Defined Functions:** You can create functions for each "room" or "scene."

22. Employee Bonus Calculator

- **Description:** Calculates an employee's bonus based on their salary and their performance rating (e.g., 1-5). A higher rating gives a larger bonus percentage.
- **Core Concepts:**
 - **If/Elif/Else:** To determine the bonus percentage based on the rating.
 - **Logical Operators:** `if rating == 5 or rating == 4:` could be used.
 - **User-Defined Functions:** `calculate_bonus(salary, rating)`.
 - **Operators:** Multiplication to calculate the bonus amount.

23. Shipping Cost Calculator

- **Description:** Calculates the shipping cost for a package based on its weight and the shipping destination (e.g., Domestic, International).
- **Core Concepts:**
 - **Nested If:** `if destination == 'domestic': if weight < 1: cost = 5 ... else: ...`
 - **Logical Statements:** Combining conditions.
 - **User-Defined Function:** `calculate_shipping(weight, destination)`.

24. Number Property Checker

- **Description:** Asks the user for an integer and tells them if it's positive/negative, even/odd, and a single-digit/multi-digit number.
- **Core Concepts:**
 - **If/Elif/Else:** To check each property independently.
 - **Nested If:** You can nest the checks, but a series of `if` statements is cleaner. A good exercise is to try both.
 - **Operators:** Modulo (%) and comparison operators (>, <).

25. Simple ATM Interface

- **Description:** Simulates a simple ATM. It asks for a PIN. If correct, it presents a menu to check balance, deposit, or withdraw. It should prevent withdrawing more than the balance.
- **Core Concepts:**
 - **Nested If:** PIN check first, then a menu inside the `if` block.
 - **If/Elif/Else:** For the main menu.
 - **Logical Statements:** To check if the withdrawal amount is valid (`amount > 0` and `amount <= balance`).
 - **User-Defined Functions:** For `check_balance`, `deposit`, `withdraw`.

26. Interactive Quiz

- **Description:** A quiz with 3-5 multiple-choice questions. It presents a question, takes the user's answer, and at the end, calculates and displays the final score.
- **Core Concepts:**
 - **If/Else:** To check if the answer is correct.
 - **User-Defined Functions:** A function `ask_question(question, options, correct_answer)` that handles the logic for a single question.
 - **Conditional Statements:** To manage the flow of the quiz.

27. Currency Converter with Multiple Currencies

- **Description:** An extension of the unit converter. It has hardcoded exchange rates (e.g., USD to EUR, USD to JPY) and allows the user to pick the 'from' and 'to' currencies.
- **Core Concepts:**
 - **Nested If:** `if from_currency == 'USD': if to_currency == 'EUR': ... elif to_currency == 'JPY': ...`
 - **User-Defined Functions:** A function for the conversion logic.
 - **String Formatting:** To display the currency symbols and formatted amounts.

28. Password Strength Checker

- **Description:** Asks the user for a password and tells them how strong it is (Weak, Medium, Strong) based on its length and whether it contains numbers, uppercase letters, and special characters.
- **Core Concepts:**
 - **String Methods:** `.isupper()`, `.islower()`, `.isdigit()`.
 - **Logical Operators:** Combining many conditions. `has_upper = any(c.isupper() for c in password)`.
 - **If/Elif/Else:** To classify the strength.
 - **User-Defined Function:** `check_strength(password)`.

29. Personal Budget Planner

- **Description:** Asks the user for their monthly income and then for their expenses in a few categories (e.g., rent, food, transport). It then calculates the total expenses and the remaining balance.
- **Core Concepts:**
 - **Variables:** To store income and each expense.
 - **Operators:** Addition and subtraction.
 - **If/Else:** To check if the user is over or under budget.
 - **User-Defined Function:** `calculate_budget(income, expenses_list)`.

30. Date Difference Calculator

- **Description:** Asks the user for two dates (as day, month, year) and calculates the number of days between them.
- **Core Concepts:**
 - **User-Defined Functions:** `is_leap_year(year)`, `days_in_month(month, year)`.
 - **Nested If:** Used within the `days_in_month` function.
 - **Complex Logical Statements:** This is a capstone project that combines nearly all the concepts you listed to solve a non-trivial problem. It's challenging but incredibly rewarding.